Ans to Ex.3

1.

$$1-(1/12)B-(1/12)B^2=0$$

$$12-B-B^2=0$$

$$B=-4 \text{ or } 3$$

Thus roots are outside unit circle and the process is stationary

Using difference equation approach, we have

$$\rho(k)=\frac{45}{77}(\frac{1}{3})^{|k|}+\frac{32}{77}\left(-\frac{1}{4}\right)^{|k|},k=0,\pm1,\pm2,...$$

2.

		k	1	2	3	4	5	6	7	8	9	10
AR parameters	(0.3,0.5)	ρ(k)	0.6	0.68	0.5	0.49	0.40	0.37	0.31	0.28	0.24	0.21
	(0.3,-0.5)		0.2	-.44	0.23	0.15	0.16	-0.027	-0.09	-0.013	0.04	0.019

3. Check (i) constant mean , (ii) constant variance, and (iii) covariance does not depend on time.

4. Since the constant 12 does not affect stationarity, we can check stationarity by looking at the process

$y_t=0.4y_{t-1}+0.1y_{t-2}$. Namely we look at the equation 1-0.4B-0.1B^2=0.

The roots are outside the unit circle, hence time series is stationary.

Then $E(y_t)=12+0.4E(y_{t-1})+0.1E(y_{t-2})$, which means

μ=12+0.4 μ+0.1 μ

and μ=24 .

5. MA(2), bookwork.

$\rho(0)=1,\rho(1)=-10/21,\rho(2)=4/21,\rho(k)=0$ for k ≥3.

6. $Z_t Z_{t-k} = 0.8 Z_{t-1} Z_{t-k} + a_t Z_{t-k} + 0.7 a_{t-1} Z_{t-k} + 0.6 a_{t-2} Z_{t-k}$ (*)

Now because Z_t has a_t as its leading term in its series expansion, and recall $\gamma(k) = E(Z_t Z_{t-k})$ $(\because E(Z_t) = 0)$, $\rho_k = \gamma_k / \gamma_0$.

Thus for $k \geq 3$, the last 3 terms in (*) has zero expectation and $\rho_k = 0.8 \rho_{k-1}$.

For k=2, $\gamma_2 = 0.8\gamma_1 + 0.6E(a_{t-2} Z_{t-2})$.

But $E(a_{t-2} Z_{t-2}) = E(a_{t-2}(0.8 Z_{t-3} + a_{t-2} + 0.7 a_{t-3} + 0.6 a_{t-4})) = \sigma_a^2$.

Thus $\rho_2 = 0.8\rho_1 + 0.6 \dfrac{\sigma_a^2}{\gamma_0}$.

Note $\rho_1 = 0.8 + 1.6 \dfrac{\sigma_a^2}{\gamma_0}$.

7. The process is $y_t = 0.3 y_{t-1} + 0.2 y_{t-2} + \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2}$ (**)

Consider $y_t = 0.3 y_{t-1} + 0.2 y_{t-2}$.

Then B= $\left| \dfrac{-3 \pm \sqrt{89}}{4} \right| > 1$. Hence time series is stationary and $E(y_t) = 0$.

Then take expectation of (**), then we have

$\rho_k = 0.3\rho_{k-1} + 0.2\rho_{k-2}$, so $\rho_3 = 0.1 \;\&\; \rho_4 = -0.01$.